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|  | <b>CSE 299: Junior Design Project</b><br><b>Department of Electrical and Computer Engineering</b> |             |                       |
|                                                                                   | <b>Semester:</b>                                                                                  | Spring 2025 | <b>Section:</b> 18    |
|                                                                                   | <b>Weekly Progress:</b>                                                                           | 1           | <b>Date:</b> 31.01.25 |

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|---------------------------------------|---------------------------------|
| <b>Faculty Name:</b>                  | Dr. Shafin Rahman               |
| <b>Group Number and Project Title</b> | Group # 7: Bank Service Chatbot |
| <b>Student Name (ID)</b>              | SUHANA ISLAM (ID: 2131550642)   |
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| Last week's Target                                                                                                                                                                              | Follow-up                     |
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| 1. The domain of our Chatbot was to be finalised.<br>2. Select the PDF on which we will be working on for our Chatbot.<br>3. Schedule a meeting date to discuss further planning of the project | 1. Done<br>2. Done<br>3. Done |

## Work narrative of the current week

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| <p>This week, I conducted thorough research to design the roadmap for our project.</p> <ul style="list-style-type: none"> <li> <b>Tasks Completed:</b> I conducted in-depth research was done on RAG(Retrieval Augmented Generation) and LLM(Large Language Model) focusing on acquiring an idea on how to implement it in on our project. </li> </ul> <p>Additionally, I researched on the key technical concepts related to our project including Ollama,Vector-Embeddings and FAISS/ ChromaDB, text chunking techniques and PDF Text extraction.</p> <p>Furthermore, a brief research was done on the Roadmap of the project. After research, I came up with a draft roadmap of the project which will include the following steps:</p> <p><b>i).Understanding RAG and Setting Up the Environment:</b></p> <ul style="list-style-type: none"> <li>- Research about RAG(Retrieval Augmented Generation) and LLM(Large Language Model).</li> <li>-Install Python and necessary packages required</li> </ul> <p><b>ii)Processing the PDF:</b></p> <ul style="list-style-type: none"> <li>-Load the PDF and extract text using ‘pymupdf’</li> <li>-Split the text into smaller,manageable chunks</li> </ul> <p><b>iii)Embedding and Storing Data:</b></p> |
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- Convert chunks into vector embeddings using Langchain
- Use Ollama for the Chatbot functionality with embeddings
- Store embeddings using FAISS or ChromaDB, vector database for retrieval.

#### iv)Building the Chatbot:

- Use retriever to extract relevant information for queries

#### v)Design the Chatbot UI:

- Create a simple UI
- Test the chatbot locally

- **Challenges faced:** Since I have not completed the majority parts of the project yet, I am still unsure about aspects of the draft roadmap might be difficult to implement.

For example,I am still deciding whether to use FAISS or ChromaDB for storing embeddings. The final decision will be made in the upcoming week after researching the pros and cons for both of them based on our project requirements.

This draft roadmap may change based on the project's requirements which are difficult to determine in advance. If any part of the project is not working smoothly as desired,I might consider switching to a different approach keeping the objectives unchanged.

- **Progress made:** I have completed Phase1 and Phase 2 according to the Roadmap.I am familiar with the RAG and LLM structure.I also setup the environment for our project along with the installation of necessary packages required for the project.I am using Visual Studio Code as the IDE for our project.

I successfully loaded our PDF and divided it into chunks having total chunks of 140.Then divided the text into chunks size of 500 having a chunk overlap of 50. So far the PDF has been successfully loaded and is divided into chunks properly.

Any changes required for the smooth running of our project will be made as I proceed into further steps.

## Target for next weekly update

For the upcoming week, I plan to complete the next phases of our project:

**1.Embedding and Storing Data:** The chunks need to be converted to vector embeddings and stored using FAISS / ChromaDB.The choice between FAISS and ChromaDB will be finalized next week.

**2.Building of the Chatbot needs to be started:** The basics of Chatbot building needs to be started such as User Handling,Processing and Understanding,Response Generation and User Interface. The implementation of these functions will take place over the following weeks, in accordance with the time management plan. It is not expected to complete all the basic structure within the next week; instead, it will be implemented gradually over time.

**3.Further Planning of Upcoming Tasks:** The following steps of the project needs to be planned in a more detailed and organized manner.

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|                                                                                   | <b>Weekly Progress:</b>                                                                           | 2           | <b>Date:</b> 14.02.25 |

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| <b>Faculty Name:</b>                  | Dr. Shafin Rahman                |
| <b>Group Number and Project Title</b> | Group # 7: Bank Customer Chatbot |
| <b>Student Name (ID)</b>              | SUHANA ISLAM (ID:2131550642)     |
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| Last week's Target                              | Follow-up |
|-------------------------------------------------|-----------|
| 4. Embedding and Storing data                   | 4. Done   |
| 5. Building of the Chatbot needs to be started. | 5. Done   |
| 6. Further Planning of Upcoming Tasks           | 6. Done   |

## Work narrative of the current week

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| <p>Here is a summary of my work on the project during the current week.</p> <ul style="list-style-type: none"> <li> <b>Tasks completed:</b> I decided to use FAISS as my vector database for this project as I was facing some challenges in using ChromaDB. So I converted the chunks into embeddings and stored them in FAISS. I installed Ollama for my project and downloaded the necessary packages required to run this project smoothly. Currently I am using llama2.         </li> </ul> <p>Before moving into User Interface, I decided to test my chatbot with a simple query before moving to UI design.</p> <p>For the User Interface, I decided to use Streamlit. Currently I just made a basic interface to check the functionality of my chatbot. The design will be modified more in the upcoming weeks. After building the basic interface, I checked the response time of my chatbot. The shortest response time recorded so far was 15 seconds.</p> <ul style="list-style-type: none"> <li> <b>Challenges Faced:</b> As I use Windows in my laptop, it doesn't have Microsoft Build Tools by default. So installing the Microsoft Build Tools will require roughly around 10GB in my Local disk(C: drive) which is currently not possible due to storage issues. Hence, using FAISS ,whose installation is a straightforward process and does not require installing additional Microsoft Build Tools, was a better option for me than ChromaDB. Moreover using FAISS will provide faster response for my project.         </li> </ul> <p>Moreover, installing Ollama was a difficult one, as by default, Ollama downloads all its packages and libraries in C drive by default and as Ollama takes up a lot of storage space, it was challenging for me as my C drive currently doesn't have enough space. Hence I had to</p> |
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change the default directory for Ollama file in my pc and change it to another hard drive which was pretty time consuming for me.

- **Progress Made:** I completed the Part 3 and 4 according to the draft Roadmap I had previously created in my previously weekly report which included embedding and storing data and starting to build the chatbot.

Before moving into UI design,I decided to test my chatbot just to make sure whether everything was working properly or not .I passed a simple query and it successfully retrieved the exact information from my pdf. For the moment, it is exactly lifting exact information from the pdf without summarization.

I checked the response time of my chatbot which is giving varied response time for different types of questions asked in the chatbot related to the pdf. Least time response recorded was 11 seconds for short questions.

I started working on Phase 5 according to my Project Roadmap Draft which was created previously which was designing the User Interface.I built a basic chatbot whose design will be further enhanced in the upcoming weeks

## Target for next weekly update

For the upcoming week,I plan to get started with the following parts of my project:

**1.Reduce the Response time of my chatbot as much as possible to get a faster reponse:** This is expected to happen gradually in the upcoming weeks rather than within just one week

**2.Build a proper User Interface for the Chatbot:** A more standard and better design will be made for the User Interface.

**3.Start researching on Automation Testing:** This chatbot will require an automation testing method rather than testing it manually every time by passing queries. A proper research on automation testing needs to be done and a brief planning should be done on how to proceed with it.

**4.Display the response time in streamlit interface:** Each time a response is generated in the chatbot by passing a query,the response time should be displayed on the streamlit interface rather than manually measuring the time taken to generate a reponse.

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|                                                                                   | <b>Weekly Progress:</b>                                  | 3           | <b>Date:</b> 27.02.25 |

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| <b>Faculty Name:</b>                  | Dr. Shafin Rahman                |
| <b>Group Number and Project Title</b> | Group # 7: Bank Customer Chatbot |
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| Last week's Target                                                                                                                                                                                                                                 | Follow-up                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 7. Reduce the response time of my chatbot as much as possible to get a faster response.<br>8. Build a proper User interface for my chatbot.<br>9. Start researching on automation testing<br>10. Display the response time on streamlit Interface. | 7. Done<br>8. Done<br>9. Done<br>10. Done |

## Work narrative of the current week

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| <p>Here is a summary of my work on the project during the current week.</p> <ul style="list-style-type: none"> <li> <b>Tasks Completed:</b> To further reduce the response time of chatbot and achieve faster responses, I tried running my project in a laptop having a NVIDIA GeForce RTX 3060 GPU. I tried running the entire project using the GPU. However, ollama is running on the CPU of that laptop but since the CPU is a powerful one, we are getting much faster response times. I set the ollama to use gpu of my laptop. Moreover, now my chatbot now paraphrases the text retrieved from PDF while generating a response, rather than direct lifting.         </li> </ul> <p>I improved the interface of my streamlit to give it a more standard, chatbot-like appearance. It now has features like storing the chats between the user and the chatbot. The time at which a query is passed, all are being noted and displayed on the page. Additionally, the interface has a light faded blue background colour. Furthermore, there is a button to clear the entire chat history.</p> <p>In order to start the testing process, I researched through the automation testing process on how to do it. It can either be done using test sets or maybe even using testing tools. The detailed planning on how to do the testing will be decided in the upcoming weeks. For the time being, we are researching all the pros and cons of using each method of testing.</p> |
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As decided in the previous week, I successfully managed to display the response time after each query is passed on my streamlit interface.

- **Challenges Faced:** I tried setting the default Ollama settings to use gpu. However, it is not using the GPU solely and is still running on the CPU of the laptop.

When I tried running my entire project in another laptop, I had to reinstall all the packages and dependencies again in that laptop and also set up a new virtual environment. Installing Ollama and Streamlit again was a challenging task for me as it is very time consuming and requires a great deal of space on C drive (local disk). Even running so many commands in my laptop, the ollama and the chatbot aren't solely running on the GPU.

- **Progress made:** I reduced the response time of my chatbot much more by some seconds by shifting my project to a laptop having a more powerful CPU and GPU. Even though the entire project isn't running solely on GPU, it is running on a much more powerful CPU resulting in a much faster response.

I built a proper chatbot interface on the streamlit adding more features to enhance its appearance further. It shows the time at which a query is passed and a response is generated. It also shows the response time taken to generate each query below the query. Currently the chat history between the user and the chatbot are being saved and displayed on the streamlit interface. It also has a button to clear the chat history.

I also researched well to get an idea on how to do automation testing for my chatbot. I decided to use test cases to test my chatbot. Further detailed planning on how to do the testing will be done in the upcoming weeks.

Initially, I will use 200-250 test sets for testing. In future, I aim to increase this to 1000 test sets.

## Target for next weekly update

For the upcoming weeks, I plan to get started on the following parts of my project:

1. **Start the Automation Testing Process:** In the upcoming weeks, I plan to at least get started with the automation testing process. It is not expected to be completed within a week; rather it will be done gradually over the upcoming weeks.
2. **Put a Reference from the pdf in my generated response:** As a response is generated, I want to include a reference from the pdf specifically the section number to indicate where the response is coming from.
3. **Further planning on how to proceed with the test sets to do the testing:** A detailed plan needs to be made on how to proceed with the automation testing process. Initially I expect the test set to be used is around 200-250 test sets. Over time, the number of test sets is expected to reach 1000.

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|                                                                                   | <b>Weekly Progress:</b>                                                                           | 4           | <b>Date:</b> 13.03.25 |

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| Last week's Target                                                                                                                                                                      | Follow-up                                                                                                                                                           |
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| 11. Start the Automation testing Process.<br>12. Put a reference from the PDF in my generated response.<br>13. Further planning on how to proceed with the test sets to do the testing. | 11. Still in progress; work has begun but the process is still not completed yet.<br>12. Still in progress; exploring ways to implement it efficiently.<br>13. Done |

## Work narrative of the current week

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| <p>In this week, I focused on both manual and automation testing for my chatbot. While manual testing was already done previous weeks, we started the automation testing but is not fully implemented yet.</p> <ul style="list-style-type: none"> <li> <b>Tasks Completed:</b> Automation testing requires a structured test set to evaluate the accuracy and efficiency of the chatbot. We are designing a test set of 1000 queries, starting with an initial batch of 200-250 test queries. The test set will contain various types of questions to ensure comprehensive testing. A draft roadmap has been created to guide the automation testing process.           </li> </ul> <p>The DRAFT ROADMAP includes the following phases:</p> <p><b>i) Creating Test Dataset:</b></p> <ul style="list-style-type: none"> <li>-Prepare a test set of 1000 test queries, initially starting with 200-250 queries.</li> <li>- Then categorize these queries into categories of short, complex and vague queries.</li> <li>-Store the expected outputs for each test query based on the chatbot's training data.</li> </ul> <p><b>ii) Implementing Testing Framework:</b></p> <ul style="list-style-type: none"> <li>-Develop a Python-based testing Script to send each test query automatically to the chatbot.</li> <li>-Capture the chatbot's responses along with the response time.</li> </ul> <p><b>iii) Analyzing Results:</b></p> |
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-After running the test set, compare the responses with the expected answers using **exact match**(for factual questions) and **semantic similarity scoring**( for complex responses).

-Identify any incorrect response and refine the chatbot's retrieval logic if needed.

#### iv)Performance Optimization:

-Analyze the response times for different query types.

-Optimize indexing, embedding retrieval and response generation for faster performance.

-Expand the test set to 1000 queries for large-scale testing.

**NOTE:** Any modifications required will be made to the draft roadmap to run the chatbot smoothly.

- **Challenges Faced:** Since chatbot responses are not deterministic, evaluating the accuracy of the responses generated is much more complex compared to traditional software testing. I faced difficulty in understanding how to verify the chatbot responses as it requires predefined expected answers which needs further research.

Running multiple queries initially was a bit challenging as it slowed down the response time. Therefore, I initially started with a small batch of 10-20 test queries. Ways to improve the response time on larger test set need to be well researched.

As I am currently working on generating the reference from the PDF, I am struggling to find an efficient way to cite the specific section of the PDF from which the chatbot retrieved information.

- **Progress Made:** Initial automation testing framework has been started. A test dataset is being created with initially 200-250 queries according to the first phase of automation testing. The test queries are being currently categorized into different categories such as short factual question, complex question, vague questions. It is still in progress and not completed yet. Manual Testing is still being conducted simultaneously to check the progress of the chatbot.

A draft Roadmap has been prepared for the upcoming weeks to guide full-scale automation testing. Any changes required to the draft roadmap will be made to run the project smoothly.

Currently, I am working on phase 1 of the roadmap. A basic automation testing script is being developed to send queries to the chatbot and evaluate the responses. I am exploring different approaches for automation testing including Python-based test execution response evaluation methods. For now, a small batch of 10-20 queries has been manually tested to evaluate the expected vs. actual chatbot responses, which helped set up the automation.

## Target for next weekly update

1. **Finish Creating the Initial 200-250 queries:** Finish preparing the initial 250 queries so that the test set can be expanded further.
2. **Optimize and Complete PDF reference retrieval:** Complete the Reference retrieval function of the chatbot. Further, enhance it to optimize it.
3. **Improve automation testing efficiency:** After testing initial test queries, use different tools to improve the efficiency of automation testing. Expand the test set further to reach a target of 1000 queries.



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|                                                                                   | <b>Weekly Progress:</b>                                  | 5           | <b>Date:</b> 28.03.25 |

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| <b>Faculty Name:</b>                  | Dr. Shafin Rahman                |
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| Last week's Target                                | Follow-up |
|---------------------------------------------------|-----------|
| 14. Finish Creating the Initial 200-250 queries   | 14.Done   |
| 15. Optimize and Complete PDF reference retrieval | 15.Done   |
| 16. Improve automation testing efficiency         | 16.Done   |

## Work narrative of the current week

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| <p>This week, I solely focused on completing the automation testing of my chatbot.</p> <ul style="list-style-type: none"> <li> <b>Tasks completed:</b> This week, I had a meeting with the RA to check if so far the work done was okay or not. After getting a positive confirmation from the RA, I proceeded with the next part of my project.         </li> </ul> <p>This week, I focused on the automation testing process. At first, I prepared an automation testing script. In the testing script, I defined scoring functions such as BLEU and ROUGE score to evaluate the chatbot's response accuracy.</p> <p>Furthermore, I prepared a CSV file for the automation testing process. Initially, I started with 10-11 tests to check whether the automation testing process was functioning properly. As I received better results, I increased my test set to 250. Later, I expanded it to 1000.</p> <p>The Streamlit interface for automation testing displays questions and answers fetched directly from the CSV file. It also displays the bot's response which helps me to compare between the expected answer and the bot's response. The BLEU and ROUGE score calculations help to evaluate the chatbot's performance. There is a button as well to stop the testing process if needed.</p> <p>The test results are displayed in a table, showing the average scores, with an option to download the CSV file. After evaluating the chatbot's performance, I observed that the BLEU and ROUGE score were low. Hence, I intend to improve this score by optimizing variables in the upcoming weeks before final project demonstration.</p> |
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- **Challenges faced:** I faced difficulty in setting up the environment for automation testing as it required preparing 1000 test sets which was pretty time consuming to find questions answers from each page of the PDF manually. Some pages of PDF contained diagrams and pie-charts, making it challenging to generate specific questions and answers.
- **Progress made:** After doing the automation testing of my chatbot, I observed low a BLEU and ROUGE scores. Hence, I am working on to improve the score to get better performance and accurate results for my chatbot.

Currently I am getting a BLEU score of 0.4124 and a ROUGE score of 0.8121. A higher BLEU and ROUGE score indicates better chatbot accuracy and performance. Therefore, in the upcoming weeks, I aim to optimize these scores to enhance the chatbot's accuracy before the final project demonstration.

## Target for next weekly update

**1.Improve BLEU and ROUGE SCORE:** Optimize the chatbot's response accuracy to achieve better performance.

**2.Prepare for final Project Demonstration**